

1. INTRODUCTION

1.1 OVERVIEW OF THE PROJECT

The Employee Leave Management System (ELMS) is a web-based application designed to automate and streamline the leave request and approval process within an organization. Traditional methods of handling leave requests often involve manual paperwork, inefficient communication, and delays, leading to decreased productivity.

ELMS addresses these challenges by offering a centralized platform for employees to apply for leave, track leave balances, and receive real-time updates, while administrators and managers can approve or reject requests efficiently.

The Employee Leave Management System (ELMS) is consist of five main modules: the Admin Module, Employee Module, Leave Management Module, Authentication & User Management Module, and Notification Module. The **Admin Module** allows administrators to manage the overall leave system, including setting leave policies, approving or rejecting leave requests, and maintaining employee leave records.

The **Employee Module** enables employees to submit leave requests, track their leave balances, and view approval statuses. The **Leave Management Module** handles the processing of leave applications, categorizing different leave types such as Casual Leave, Sick Leave, Earned Leave, and Restricted Holidays (RH) while ensuring compliance with leave policies. The **Authentication & User Management Module** manages user access, ensuring secure login credentials for employees and administrators while maintaining role-based access control. Lastly, the **Notification Module** automates the communication process by sending email notifications to employees and administrators.

The **Employee Leave Management System (ELMS)** supports two types of users: **admin and employee**. The **admin** is responsible for managing employee leave requests, setting leave policies, tracking leave balances, generating reports, and approving or rejecting leave applications. The **employee** can log in to submit leave requests, check leave balances, and track the status of their applications. If approved or rejected, employees receive automatic notifications via email.

Employee leave management system combine number of processes and systems to automate and easily manage employee data, leave request, track and grant leave. In many institution staff are entitled to different types of leave, these leave are granted according to institution policy. Administrative department is mostly responsible for managing and granting leave request. To this end, most institution used conventional method of requesting, granting and managing leave.

In conventional method, leave is manually request by writing letter to head of department. The head of department minutes and forward the request to higher staff for approval. This method is time consuming, prone to error, require more paper work and difficult to manage. Hence the need for an automated leave management system that is faster, error free, less paper work and easy to manage.

The system was achieved by developing an automated employee leave management system using the three tier software architectural model. The System is implemented using web based technologies which include CSS, JS, HTML, MySQL, PHP and runs on Windows operating system. . The overall functionality of the system shows that it work satisfactory and the result obtained shows that the system is error free, faster and allows staff to request for leave in a timely manner. Hence the system can be used by both academic staff and administrative department of an institution for effective and efficient management of employee leave.

The traditional manual approach to employee leave management often leads to inefficiencies, errors, and difficulties in tracking and managing leave requests. This paper presents the development and implementation of an automated Employee Leave Management System (ELMS) designed to address these challenges. The ELMS provides a centralized, user-friendly platform for employees to submit leave requests, managers to approve or reject them, and HR to track leave balances and policies. By automating tasks such as leave request submission, approval workflows, and leave balance tracking, the ELMS aims to improve efficiency, reduce errors, and enhance the overall management of employee leave within the organization. The system's key features include: a secure login system for employees and managers, customizable leave types and policies, automated notifications for pending requests, and comprehensive reporting capabilities.

The implementation of the ELMS has resulted in a significant reduction in paperwork, improved accuracy in leave tracking, and enhanced communication between employees,

managers, and HR. The system's user-friendly interface and streamlined processes have also contributed to increased employee satisfaction and a more efficient HR department. The ELMS serves as a valuable tool for organizations seeking to optimize their leave management processes and improve overall workforce management.

Employee Leave Management System is a distributed application, developed to maintain the details of employees working in any organization. It maintains the information about the personal details of their employees, also the details about the leave status. The application is developed using PHP. It is simple to understand and can be used by anyone who is not even familiar with simple employees' system. It is user friendly and just asks the user to follow step by step operations by giving him/her few options.

It is fast and can perform many operations of a company. This software package has been developed using the HTML, CSS, JavaScript and bootstrap at Front End and My SQL Server and PHP at Back End. The software is very user friendly. The package contains different modules like Employee, Admin and Manager.

Employee Leave Management System Project is designed to keep track of employee leave information in any company. It stores data such as their employees' personal information, leave details and salary details. The employee leave management system project gives managers a better idea of their employees and helps them plan and manage their work hours to cut costs and boost productivity. It gives appropriate directions and supervisions for employees. It also secures and manages information that are important to the employees including personal and work-related information.

Employee leave management system is developed to override the problems prevailing in the practicing manual system. It allows employee and manager to get clear insight into the employee leave history. The detailed leave history will help an employee to plan for their future leave.

1.2 ORGANIZATION PROFILE

The Employee Leave Management System (ELMS) is designed to streamline leave application, tracking, and approval processes for organizations. The system ensures transparency and efficiency in managing employee leave requests while maintaining accurate records of leave balances.

To ensure the effectiveness of ELMS, data was collected from various corporate offices and HR departments to understand leave policies and management challenges. The system is built to accommodate different leave types, employee roles, and approval hierarchies, making it suitable for organizations of all sizes.

ELMS enforces strict leave policies, maintaining a structured leave distribution system. The total annual leave limit is 24 days, categorized into Casual Leave (12 days), Sick Leave (5 days), Earned Leave (4 days), and Restricted Leave (3 days). This ensures transparency and prevents misuse of leave entitlements. Employees can track the status of their leave requests in real-time and receive notifications regarding approval or rejection.

An Employee Leave Management System (ELMS) is a software application that helps organizations manage employee leave requests, track leave balances, and ensure compliance with leave policies. Below is an outline of a comprehensive ELMS for an organization

The ELMS enables employees to request leaves, managers to approve/reject them, and HR to maintain records. It ensures transparency, compliance, and efficient workforce planning

It is difficult to provide an individual leave report, as well as there will be complicated in searching, editing, and updating the data. Since it is complicated in searching, editing, and updating it will consume lots of time. So, it is time consuming process as well. Employee is not able to see any other employees leave status and it is difficult to manage and track multiple type of an employee leave.

1.3PROBLEM DEFINITION AND OBJECTIVES

1.3.1 PROBLEM DEFINITION

Managing employee leave manually or using traditional methods such as paper forms, emails, or spreadsheets creates several inefficiencies in an organization. The process often leads to delays in leave approvals, miscommunication between employees and managers, and difficulty in tracking leave balances accurately.

Without a centralized system, HR departments face challenges in maintaining proper records, ensuring compliance with company leave policies, and managing overlapping leave requests, which can disrupt workflow and productivity. Additionally, employees may struggle to access real-time information about their leave status, leading to confusion and repeated inquiries.

The lack of automation also increases the risk of errors in payroll processing, particularly when managing paid and unpaid leaves. To address these challenges, an **Employee Leave Management System (ELMS)** is required to automate and streamline the leave request, approval, and tracking process. By implementing a digital solution, organizations can improve efficiency, enhance transparency, and ensure seamless communication between employees, managers, and HR teams, ultimately leading to better workforce management.

The use of paperwork in handling some of the processes could lead to human error, papers may end up in the wrong hands and not forgetting the facts that this is time consuming. A few current systems lack employee self-service meaning employees are not able to access and manage their personal information directly without having to go through their HR department or their managers. Another challenge is that multi-national companies will have all the employee information stored at the headquarters of the company making it difficult to access the employee information from remote place when needed at short notice.

The project is aimed at setting up an employee's information system about the status of the employee, the educational background, and the work experience to help monitor the performance and achievement of the employee through password protected system.

1.3.2 OBJECTIVES

- ❖ Provide an easy-to-use platform for employees to apply for leave.
- ❖ Speed up the leave approval process by reducing manual work.
- ❖ Minimize paperwork and administrative overhead for HR teams.
- ❖ Maintain real-time leave balances for each employee.
- ❖ Notify employees instantly about leave approvals or rejections.
- ❖ Allow admins to approve or reject requests quickly.
- ❖ Make it easy for employees to check their leave status anytime.

2. SYSTEM STUDY AND ANALYSIS

2.1 EXISTING SYSTEM

The existing system is a manual one. It is a difficult job to maintain the record for the user. The human effort is more. In many organizations, leave management is handled manually or through basic spreadsheets. Employees typically apply for leave using paper forms which HR or managers process manually. The existing system is not user friendly because the retrieval of data is time-consuming and data is not maintained efficiently.

Following are the steps taken during the initial study: Initially, we collected all the information, which they wanted to store. Then we studied the working of the current system which is done manually. We noted the limitation of that system which motivated them to have new system. With the help of these documents we got basic ideas about the system as well as input output of the developed system. The most important thing is to study system thoroughly.

Here we are studying both existing system and proposed system so that advantages & disadvantages of both the systems can be understood. The first task was identifying how system can be computerized. Some analysis and projections was done regarding changes to be made to the existing system

2.1.1 DISADVANTAGES OF EXISTING SYSTEM

- Traditional leave management systems often suffer from several challenges that impact efficiency. **Manual errors** are common when tracking leave balances and approvals, leading to discrepancies in records.
- The process is also **time-consuming**, requiring HR personnel to manually verify leave requests and update records. Additionally, there is a **lack of real-time tracking**, making it difficult for employees and admins to monitor leave balances and application statuses accurately.
- Organizations also face **difficulty in enforcing leave policies**, as manual systems struggle to restrict leave applications based on predefined rules..
- Without a notification **system**, employees and managers remain unaware of leave approvals, rejections, or pending requests. The system also lacks **comprehensive**

reporting and analytics, making it difficult for HR to analyze leave trends and maintain workforce efficiency.

- Lastly, managing multiple leave types becomes complicated, as each category has its own rules and limits, increasing the chances of mismanagement and policy violations. An automated Employee Leave Management System effectively addresses these issues, improving accuracy, efficiency, and compliance.

2.2 PROPOSED SYSTEM

The proposed system of the Leave Management System reduces the over work and time duration. It provides an easy, maintained and systematic environment for the higher panel (HR/Manager) for the approval of leave. The proposed system chances of data losing is not possible because data is maintained in the form of database and digitalization of Employee's information. This application is an online which makes more flexible to access information.

System In proposed system chances of losing data is not possible because data is maintained in the form of database. Every day attendance, leaves and notices information is updated in to database using a user friendly GUI. This will reduce work for Head of departments. This application is an online application which makes more flexible to access information

2.2.1 ADVANTAGES OF PROPOSED SYSTEM

- The **Employee Leave Management System (ELMS)** streamlines the leave management process by **reducing manual work and minimizing errors** in leave calculations. Being a web-based system, it **can be accessed globally with the help of the internet**, allowing employees to apply for leave from anywhere at any time.
- Employees can also **instantly check their leave balance and status**, ensuring transparency and convenience.
- Additionally, both employees and administrators can **view leave history and approvals**, making tracking and managing leave requests more efficient. Since all details are **securely stored in a centralized database**, maintaining records becomes easier, eliminating paperwork and data loss.

- The system also **simplifies online leave management and servicing**, offering a user-friendly experience. Moreover, it ensures **secure database management**, protecting employee leave records from unauthorized access and data breaches.

2.3 FEASIBILITY ANALYSIS

A feasibility Study is a Comprehensive evaluation of a proposed project that evaluates, all factors critical to its Success to assess its likelihood of success. A feasibility study evaluates a proposed plan or practicality and Viability to determine its Success. As the name implies, a feasibility analysis is used to determine the viability of an idea such as ensuing a project is regally, technically and economically justifiable.

1. Technical Feasibility

The **Employee Leave Management System (ELMS)** utilizes widely available technologies such as PHP, MySQL, and web development frameworks for smooth functionality. The system integrates a database management system (**MySQL**) to efficiently store and manage employee leave records. Additionally, the system includes email notifications for leave approval updates and reminders.

2. Operational Feasibility

The System is designed to be user-friendly for employees, HR personnel, and administrators. The system automates key processes such as leave application, approval workflow, and leave balance tracking, reducing manual effort and minimizing human errors. A centralized database ensures quick and reliable access to employee leave records. The system also generates graphical reports. Email notifications for approvals and rejections, employees receive real-time updates on their leave status.

3. Economic Feasibility

The initial investment in software and hardware is reasonable, aligning with the project's scope. automating **leave requests, approvals, and record-keeping**, the system significantly **reduces costs** associated with manual paperwork, printing, and maintaining physical records.

2.4 SYSTEM SPECIFICATION

The Employee Leave Management System (ELMS) is designed to help employees apply for leave and managers/admins to track, approve, and manage leave requests efficiently. The system ensures compliance with leave policies and provides real-time leave balance tracking. This specification outlines the hardware and software requirements essential for the development and implementation of the system.

2.4.1 HARDWARE SPECIFICATION

Hardware specification refers to the minimum and recommended physical components required to develop and run project efficiently. It includes details about hardware components like processor, memory (RAM) and storage. Hardware requirements for the present project is,

- ❖ Processor : Intel Core i5
- ❖ RAM : 4GB
- ❖ Storage : 250GB

2.4.2 SOFTWARE SPECIFICATION

Software specification refers to the required software tools, technologies and programming languages needed to develop, test, and run the project efficiently.

- ❖ Operating System : Windows 11
- ❖ Backend : PHP
- ❖ Frontend : HTML5, CSS3, JavaScript
- ❖ Database : MySQL
- ❖ Web Browser : Google Chrome
- ❖ Editor : Visual Studio Code

2.4.2.1 FRONTEND

HTML (Hyper Text Markup Language)

How is it used in this project?

HTML (HyperText Markup Language) is essential for building the Employee Leave Management System (ELMS) as it provides the structure and layout of the system's web pages. It helps create important sections like the login page, dashboard, leave application form, and leave history page, making it easy for employees and administrators to use the system. HTML forms allow users to apply for leave, select leave types, and submit requests using simple elements like text fields, dropdown menus, and buttons.

Tables are used to display leave records in an organized way, showing employee details, leave dates, types, and approval status. Navigation menus and buttons help users move between different sections, such as leave requests, leave balances, and approvals. HTML also works with CSS to improve the design and JavaScript to add interactive features like form validation and date selection. Overall, HTML acts as the backbone of the ELMS, making it easy to use, well-organized, and efficient for managing employee leave.

Web browsers receive HTML documents from a web server or from local storage and render the documents into multimedia web pages. HTML describes the structure of a web page semantically and originally included cues for its appearance.

HTML elements are the building blocks of HTML pages. With HTML constructs, images and other objects such as interactive forms may be embedded into the rendered page. HTML provides a means to create structured documents by denoting structural semantics for text such as headings, paragraphs, lists, links, quotes, and other items. HTML elements are delineated by tags, written using angle brackets. Tags such as and <input> directly introduce content into the page. Other tags such as <p> and </p> surround and provide information about document text and may include sub-element tags. Browsers do not display the HTML tags but use them to interpret the content of the page.

HTML can embed programs written in a scripting language such as JavaScript, which affects the behavior and content of web pages. The inclusion of CSS defines the look and layout of content. The World Wide Web Consortium (W3C), former maintainer of the HTML and current maintainer of the CSS standards, has encouraged the use of CSS over explicit presentational HTML since 1997.[3] A form of HTML, known as HTML5, is used to display video and audio, primarily using the <canvas> element, together with JavaScript.

HTML (HyperText Markup Language) is the most basic building block of the Web. It defines the meaning and structure of web content. Other technologies besides HTML are generally used to describe a web page's appearance/presentation (CSS) or functionality/behavior (JavaScript).

"Hypertext" refers to links that connect web pages to one another, either within a single website or between websites. Links are a fundamental aspect of the Web. By uploading content to the Internet and linking it to pages created by other people, you become an active participant in the World Wide Web.

HTML uses "markup" to annotate text, images, and other content for display in a Web browser. HTML markup includes special "elements" such as <head>, <title>, <body>, <header>, <footer>, <article>, <section>, <p>, <div>, , , <aside>, <audio>, <canvas>, <datalist>, <details>, <embed>, <nav>, <search>, <output>, <progress>, <video>, , , and many others.

An HTML element is set off from other text in a document by "tags", which consist of the element name surrounded by < and >. The name of an element inside a tag is case-insensitive. That is, it can be written in uppercase, lowercase, or a mixture. For example, the <title> tag can be written as <Title>, <TITLE>, or in any other way. However, the convention and recommended practice is to write tags in lowercase.

CSS (Cascading Style Sheets)

How is it used in this project?

CSS (Cascading Style Sheets) is important in the Employee Leave Management System (ELMS) because it makes the system look better, easier to use, and work well on different devices. It helps design key pages like the login page, dashboard, and leave application form, making them attractive and user-friendly. With responsive design, the system adjusts to different screen sizes, so employees can use it on desktops, tablets, or mobile phones without any issues. CSS also improves tables that show leave records by adding borders, different row colors, and hover effects to make the data clearer.

Buttons and forms are styled with hover effects, rounded edges, and shadows, making them smooth and modern. It also allows for theme customization, so companies can match the system's colors to their brand, and even add a dark mode option. CSS adds animations and transitions like smooth pop-ups and sliding menus to improve the user experience. Overall, CSS makes ELMS visually appealing, responsive, and easy to use, ensuring a professional and engaging system for employees and administrators.

CSS is designed to enable the separation of content and presentation, including layout, colors, and fonts.[4] This separation can improve content accessibility, since the content can be written without concern for its presentation; provide more flexibility and control in the specification of presentation characteristics; enable multiple web pages to share formatting by specifying the relevant CSS in a separate .css file, which reduces complexity and repetition in the structural content; and enable the .css file to be cached to improve the page load speed between the pages that share the file and its formatting.

Separation of formatting and content also makes it feasible to present the same markup page in different styles for different rendering methods, such as on-screen, in print, by voice (via speech-based browser or screen reader), and on Braille-based tactile devices. CSS also has rules for alternative formatting if the content is accessed on a mobile device.

The name cascading comes from the specified priority scheme to determine which declaration applies if more than one declaration of a property match a particular element. This cascading priority scheme is predictable.

The CSS specifications are maintained by the World Wide Web Consortium (W3C). Internet media type (MIME type) text/css is registered for use with CSS by RFC 2318 (March 1998). The W3C operates a free CSS validation service for CSS documents.

In addition to HTML, other markup languages support the use of CSS including XHTML, plain XML, SVG, and XUL. CSS is also used in the GTK widget toolkit.

Each web browser uses a layout engine to render web pages, and support for CSS functionality is not consistent between them. Because browsers do not parse CSS perfectly, multiple coding techniques have been developed to target specific browsers with workarounds (commonly known as CSS hacks or CSS filters).

The adoption of new functionality in CSS can be hindered by a lack of support in major browsers. For example, Internet Explorer was slow to add support for many CSS 3 features, which slowed the adoption of those features and damaged the browser's reputation among developers. Additionally, a proprietary syntax for the non-vendor-prefixed filter property was used in some versions. In order to ensure a consistent experience for their users, web developers often test their sites across multiple operating systems, browsers, and browser versions, increasing development time and complexity.

Tools such as BrowserStack have been built to reduce the complexity of maintaining these environments. In addition to these testing tools, many sites maintain lists of browser support for specific CSS properties, including CanIUse and the MDN Web Docs. Additionally, CSS 3 defines feature queries, which provide an @supports directive that will allow developers to target browsers with support for certain functionality directly within their CSS.

CSS that is not supported by older browsers can also sometimes be patched in using JavaScript polyfills, which are pieces of JavaScript code designed to make browsers behave consistently. These workarounds—and the need to support fallback functionality—can add complexity to development projects, and consequently, companies frequently define a list of browser versions that they will and will not support.

As websites adopt newer code standards that are incompatible with older browsers, these browsers can be cut off from accessing many of the resources on the web (sometimes intentionally). Many of the most popular sites on the internet are not just visually degraded on older browsers due to poor CSS support but do not work at all, in large part due to the evolution of JavaScript and other web technologies.

JavaScript

How is it used in this project?

JavaScript is an important part of the Employee Leave Management System (ELMS) because it makes the system more interactive, user-friendly, and efficient. It helps validate forms by checking if all required fields are filled, ensuring employees do not apply for more leave than available, and making sure the dates are correct. JavaScript also updates leave balances in real-time, so employees can see how many days they have left before submitting a request.

JavaScript allows the system to fetch and update data without refreshing the page, making leave history and status updates faster. It also enables date pickers and calendars, allowing employees to select leave dates easily while blocking past dates and highlighting holidays. Additionally, JavaScript sends notifications and alerts for important updates, such as leave approvals, upcoming leaves, or errors in applications. It also adds animations and effects like smooth transitions, loading spinners, and highlighted messages to improve the user experience. Overall, JavaScript plays a key role in making ELMS faster, easier to use, and more efficient for both employees and administrators.

Web browsers have a dedicated JavaScript engine that executes the client code. These engines are also utilized in some servers and a variety of apps. The most popular runtime system for non-browser usage is Node.js.

JavaScript is a high-level, often just-in-time compiled language that conforms to the ECMAScript standard. It has dynamic typing, prototype-based object-orientation, and first-class functions. It is multi-paradigm, supporting event-driven, functional, and imperative programming styles. It has application programming interfaces (APIs) for working with text, dates, regular expressions, standard data structures, and the Document Object Model (DOM).

The ECMAScript standard does not include any input/output (I/O), such as networking, storage, or graphics facilities. In practice, the web browser or other runtime system provides JavaScript APIs for I/O.

Although Java and JavaScript are similar in name and syntax, the two languages are distinct and differ greatly in design.

A JavaScript engine is a software component that executes JavaScript code. The first JavaScript engines were mere interpreters, but all relevant modern engines use just-in-time compilation for improved performance. JavaScript engines are typically developed by web browser vendors, and every major browser has one. In a browser, the JavaScript engine runs in concert with the rendering engine via the Document Object Model and Web IDL bindings.

However, the use of JavaScript engines is not limited to browsers; for example, the V8 engine is a core component of the Node.js runtime system. Since ECMAScript is the standardized specification of JavaScript, ECMAScript engine is another name for these implementations. With the advent of WebAssembly, some engines can also execute this code in the same sandbox as regular JavaScript code.

A JavaScript engine must be embedded within a runtime system (such as a web browser or a standalone system) to enable scripts to interact with the broader environment. The runtime system includes the necessary APIs for input/output operations, such as networking, storage, and graphics, and provides the ability to import scripts.

JavaScript is a single-threaded language. The runtime processes messages from a queue one at a time, and it calls a function associated with each new message, creating a call stack frame with the function's arguments and local variables. The call stack shrinks and grows based on the function's needs. When the call stack is empty upon function completion, JavaScript proceeds to the next message in the queue.

This is called the event loop, described as "run to completion" because each message is fully processed before the next message is considered. However, the language's concurrency model describes the event loop as non-blocking: program I/O is performed using events and callback functions. This means, for example, that JavaScript can process a mouse click while waiting for a database query to return information.

2.4.2.2 BACKEND

PHP (Hypertext Preprocessor)

How is it used in this project?

PHP plays a key role in the Employee Leave Management System (ELMS) by handling the backend operations and connecting the system to the database. It processes leave applications, user logins, and approvals while securely storing data. When an employee applies for leave, PHP checks the leave balance, updates records, and sends notifications to administrators. It also retrieves and displays leave history, ensuring employees and admins can track past and pending requests.

PHP helps in user authentication, making sure only authorized users can access the system. With PHP, the system can handle real-time leave calculations, preventing employees from applying for more leave than allowed. It also supports email notifications to inform employees about leave status updates. Overall, PHP acts as the brain of ELMS, managing data, processing requests, and ensuring smooth communication between users and the database.

PHP code is usually processed on a web server by a PHP interpreter implemented as a module, a daemon or a Common Gateway Interface (CGI) executable. On a web server, the result of the interpreted and executed PHP code—which may be any type of data, such as generated HTML or binary image data—would form the whole or part of an HTTP response. Various web template systems, web content management systems, and web frameworks exist that can be employed to orchestrate or facilitate the generation of that response. Additionally, PHP can be used for many programming tasks outside the web context, such as standalone graphical applications and drone control. PHP code can also be directly executed from the command line.

The standard PHP interpreter, powered by the Zend Engine, is free software released under the PHP License. PHP has been widely ported and can be deployed on most web servers on a variety of operating systems and platforms.

The PHP language has evolved without a written formal specification or standard, with the original implementation acting as the de facto standard that other implementations aimed to follow.

W3Techs reports that as of 27 October 2024 (about two years since PHP 7 was discontinued and 11 months after the PHP 8.3 release), PHP 7 is still used by 50.0% of PHP websites, which is outdated and known to be insecure.[18][19] In addition, 13.2% of PHP websites use the even more outdated (discontinued for 5+ years) and insecure PHP 5, and the no longer supported PHP 8.0 is also very popular, so the majority of PHP websites do not use supported versions.

The only complete PHP implementation is the original, known simply as PHP. It is the most widely used and is powered by the Zend Engine. To disambiguate it from other implementations, it is sometimes unofficially called "Zend PHP". The Zend Engine compiles PHP source code on-the-fly into an internal format that it can execute, thus it works as an interpreter. It is also the "reference implementation" of PHP, as PHP has no formal specification, and so the semantics of Zend PHP define the semantics of PHP. Due to the complex and nuanced semantics of PHP, defined by how Zend works, it is difficult for competing implementations to offer complete compatibility.

PHP's single-request-per-script-execution model, and the fact that the Zend Engine is an interpreter, leads to inefficiency; as a result, various products have been developed to help improve PHP performance. In order to speed up execution time and not have to compile the PHP source code every time the web page is accessed, PHP scripts can also be deployed in the PHP engine's internal format by using an opcode cache, which works by caching the compiled form of a PHP script (opcodes) in shared memory to avoid the overhead of parsing and compiling the code every time the script runs. An opcode cache, Zend OpCache, is built into PHP since version 5.5. Another example of a widely used opcode cache is the Alternative PHP Cache (APC), which is available as a PECL extension.

While Zend PHP is still the most popular implementation, several other implementations have been developed. Some of these are compilers or support JIT compilation, and hence offer performance benefits over Zend PHP at the expense of lacking full PHP compatibility.[citation needed] Alternative implementations include the following:

HHVM (HipHop Virtual Machine) – developed at Facebook and available as open source, it converts PHP code into a high-level bytecode (commonly known as an intermediate language), which is then translated into x86-64 machine code dynamically at runtime by a just-in-time (JIT) compiler, resulting in up to 6× performance improvements.

However, since version 7.2 Zend has outperformed HHVM,[240] and HHVM 3.24 is the last version to officially support PHP.

HipHop – developed at Facebook and available as open source, it transforms the PHP scripts into C++ code and then compiles the resulting code, reducing the server load up to 50%. In early 2013, Facebook deprecated it in favour of HHVM due to multiple reasons, including deployment difficulties and lack of support for the whole PHP language, including the `create_function()` and `eval()` constructs.

Parrot – a virtual machine designed to run dynamic languages efficiently; the cross-compiler Pipp transforms the PHP source code into the Parrot intermediate representation, which is then translated into the Parrot's bytecode and executed by the virtual machine.

PeachPie – a second-generation compiler to .NET Common Intermediate Language (CIL) bytecode, built on the Roslyn platform; successor of Phalanger, sharing several architectural components

Phalanger – compiles PHP into .Net Common Intermediate Language bytecode; predecessor of PeachPie
Quercus – compiles PHP into Java bytecode

MySQL

How is it used in this project?

MySQL is used in the Employee Leave Management System (ELMS) to store and manage all data securely. It keeps records of employees, leave requests, leave balances, approvals, and rejections in a structured way. When an employee applies for leave, MySQL stores the request in the database and updates the available leave balance. It also helps in retrieving past leave history, so employees and administrators can track approved, pending, and rejected leaves. MySQL ensures that leave policies are followed by storing different leave limits and preventing excess leave applications.

It also helps in user authentication, making sure only authorized users can access the system. Overall, MySQL works as the database backbone of ELMS, ensuring secure, fast, and organized data management for smooth leave processing.

MySQL is free and open-source software under the terms of the GNU General Public License, and is also available under a variety of proprietary licenses. MySQL was owned and sponsored by the Swedish company MySQL AB, which was bought by Sun Microsystems (now Oracle Corporation). In 2010, when Oracle acquired Sun, Widenius forked the open-source MySQL project to create MariaDB.

MySQL has stand-alone clients that allow users to interact directly with a MySQL database using SQL, but more often, MySQL is used with other programs to implement applications that need relational database capability. MySQL is a component of the LAMP web application software stack (and others), which is an acronym for Linux, Apache, MySQL, Perl/PHP/Python. MySQL is used by many database-driven web applications, including Drupal, Joomla, phpBB, and WordPress.

The MySQL server software itself and the client libraries use dual-licensing distribution. They are offered under GPL version 2, or a proprietary license.

Support can be obtained from the official manual. Free support additionally is available in different IRC channels and forums. Oracle offers paid support via its MySQL Enterprise products. They differ in the scope of services and in price. Additionally, a number of third party organisations exist to provide support and services.

MySQL has received positive reviews, and reviewers noticed it "performs extremely well in the average case" and that the "developer interfaces are there, and the documentation (not to mention feedback in the real world via Web sites and the like) is very, very good". It has also been tested to be a "fast, stable and true multi-user, multi-threaded SQL database server".

SERVER SIDE

XAMPP Server

How is it used in this project?

XAMPP is used in the Employee Leave Management System (ELMS) as a local server to run the project on a computer. It includes Apache, MySQL, PHP, and phpMyAdmin, which help in developing and testing the system before hosting it online. Apache acts as the web server, allowing the system to work in a browser. MySQL stores all employee and leave data securely, while PHP processes leave applications and approvals.

phpMyAdmin helps in managing the database easily. With XAMPP, developers can test the system locally, check for errors, and make improvements before making it live. Overall, XAMPP provides a complete environment to develop, run, and manage the ELMS smoothly on a personal computer.

XAMPP is a free and open-source cross-platform web server solution stack package developed by Apache Friends, consisting mainly of the Apache HTTP Server, MariaDB database, and interpreters for scripts written in the PHP and Perl programming languages. Since most actual web server deployments use the same components as XAMPP, it makes transitioning from a local test server to a live server possible.

XAMPP's ease of deployment means a WAMP or LAMP stack can be installed quickly and simply on an operating system by a developer, with the advantage that common add-in applications such as WordPress and Joomla! can also be installed with similar ease using Bitnami.

XAMPP is regularly updated to the latest releases of Apache, MariaDB, PHP and Perl. It also comes with a number of other modules, including OpenSSL, phpMyAdmin, MediaWiki, Joomla, WordPress and more. Self-contained, multiple instances of XAMPP can exist on a single computer, and any given instance can be copied from one computer to another.

XAMPP is offered in both a full and a standard version (Smaller version).The most obvious characteristic of XAMPP is the ease at which a WAMP webserver stack can be deployed and instantiated. Later, some common packaged applications that could be easily installed were provided by Bitnami.

Officially, XAMPP's designers intended it for use only as a development tool, to allow website designers and programmers to test their work on their own computers without any access to the Internet. To make this as easy as possible, many important security features are disabled by default. XAMPP has the ability to serve web pages on the World Wide Web. A special tool is provided to password-protect the most important parts of the package.

XAMPP also provides support for creating and manipulating databases in MariaDB and SQLite, among others.Once XAMPP is installed, it is possible to treat a local host like a remote host by connecting using an FTP client. Using a program like FileZilla has many advantages when installing a content management system (CMS) like Joomla or WordPress[[further explanation needed](#)]. It is also possible to connect to localhost via FTP with an HTML editor.

3. SYSTEM DESIGN AND DEVELOPMENT

3.1 ER DIAGRAM

An Entity-Relationship (ER) Diagram is a visual representation of a database structure that shows the relationships between different entities in a system. It is used in database design to model how data is stored and linked together.

Components of ER Diagram

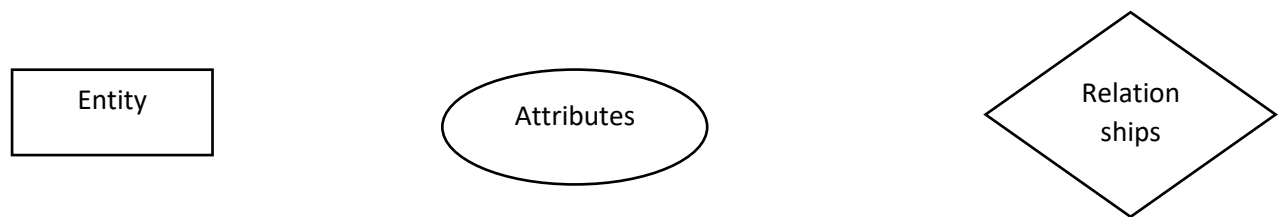


Figure 3.1.1

- **Entity** - Entities are source and destination of information data. Entities are represented by rectangles with their respective names.
- **Attributes** - define the properties or characteristics of an entity. They are represented by ovals in an ER diagram and are connected to their respective entities. Each entity consists of multiple attributes that provide detailed information about it.
- **Relationships** - define how two or more entities are connected in an ER diagram. They are represented by diamonds with their respective relationship names.

ER diagram

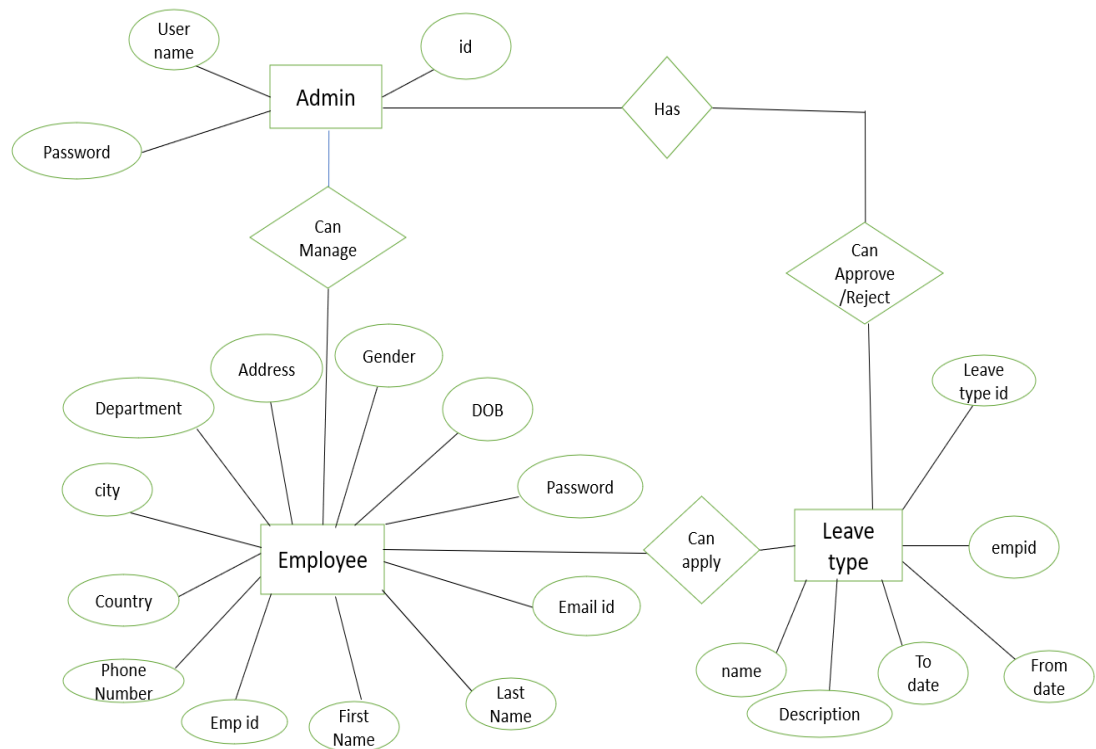


Figure 3.1.2

3.2 INPUT DESIGN

Input design refers to the process of defining how leave-related data is collected, entered, and processed within the system. It ensures accurate, efficient, and user-friendly data entry for employees, and administrators. The input design converts user-entered leave requests into system-processed records.

1. **Login Form (Figure c.1):** Users enter their **Employee id, Email, password**, and select their **role** (Admin/Employee). Validation ensures correct credentials before granting access.
2. **Employee Registration (Figure c.6):** Employees provide details like name, email, phone number, department, designation, and joining date. Leave balance is auto-calculated, and the password must meet security criteria.
3. **Leave Application (Figure c.3):** Employees select leave type, duration, and reason, with an option to upload supporting documents. The system checks leave balance and prevents past date applications.
4. **Leave Approval (Figure c.8):** Admins review leave requests with details like employee name, leave type, and reason. They can approve/reject requests, with remarks required for rejections.
5. **Leave Balance & Reports (Figure c.2):** Employees/admins can view leave balance and generate reports by filtering employee ID, department, leave type, or date range.

3.3 OUTPUT DESIGN

Output design refers to the results and information generated by the system for various end users, such as employees, and administrators. Outputs are based on user inputs and system processing, ensuring that information is presented in a clear and meaningful way.

1. Dashboard Display (Figure d.1):

- After login, users see a summary of pending leave requests, approved leaves, rejected leaves, and leave balances.
- Admins see an overview of total employees, leave trends, and system notifications.

2. Leave Application Status (Figure d.2):

- Employees receive instant notifications on their leave request status (Approved/Pending/Rejected) with admin remarks.
- The system generates error messages if leave rules are violated (e.g., exceeding leave balance).

3. Employee Leave Balance Statement (Figure d.3):

- Employees can view their remaining leave balance categorized by Casual Leave, Sick Leave, and Annual Leave.
- The system updates leave balances in real-time after each approved leave request.

4. Admin Review Panel (Figure d.3):

- Admins can view, approve, or reject leave requests with detailed employee leave history.

3.4 DATABASE DESIGN

The proposed system consists of 5 tables in the database. They are

- **Admin** (Table a.1)

Stores Admin credentials like, id, username, password.

PRIMARY KEY: id

- **Leave count** (Table a.2)

Stores Leave count details like, Leave count_id, employee_id, leave count, creates_at, updated_at.

PRIMARY KEY: leavecount_id

FOREIGN KEY: employee_id

- **Employee** (Table a.3)

Stores employee details like, employee_id, firstname, lastname, email, password, gender, Dob, department, address, city, country, phonenumber, regdate.

PRIMARY KEY: employee_id

- **Leavetype** (Table a.4)

Stores leave type details like, leavetype_id, Casual leaves, Sick leaves, Earned leaves, Restricted leaves.

PRIMARY KEY: leavetype_id

- **Leave Request** (Table a.5)

Stores Leave request details like, Leaverequest_id, employee_id, name, leavetype, from date, to date, posting date, admin remark, admin remark date, leave count, status, isread.

PRIMARY KEY: leavetype_id

FOREIGN KEY: employee_id

3.5 MODULE DESIGN

Descriptions of the Module

1. Admin Module

This module allows admin can login securely. Admin can login by entering username and password. The admin module is responsible for managing employees, leave types. It allows administrators to approve or reject leave requests based on company policies.

2. Employee Module

This module allows employee can login securely. Employee can login entering Email Id, password. Employees can easily apply for leave through the system, specifying the type and duration of leave. They can also view their leave status and remaining leave balance to track their usage. Additionally, the system allows employees to check their leave history.

3. Leave Management Module

The system defines various leave types, such as Casual, Sick, Sick leave, Earned leave, Restricted leave, with specific limits for each category. It enforces leave restrictions, ensuring that employees do not exceed the maximum allowed leaves, such as 24 total leaves per year.

4. Reports & Notifications Module

The system automatically sends email notifications to employees regarding the approval or rejection of their leave requests.

5. Leave Balance Module

Employees can check their available leave balance. When a leave request is approved, the system updates the leave balance accordingly.

4. SYSTEM IMPLEMENTATION

System implementation refers to the process of designing, developing, testing, and deploying a system to ensure it functions as expected. In the project Employee Leave Management System (ELMS), system implementation involves creating a web-based platform that enables employees to apply for leave, track leave balances, and allows administrators to manage leave requests effectively.

The Employee Leave Management System is developed using PHP, MySQL, HTML, CSS, and JavaScript to efficiently manage leave requests and approvals. The system consists of two main user roles: Admin and Employee. The Admin manages employee details, leave types, leave requests, and company leave policies, while Employees can apply for leave, view their leave balances, and track the status of their requests.

The database is structured with tables for admin, employee, leavecount, leaverequest, leave types to ensure proper data organization and efficient management of leave records.

USER MANUAL

The Employee Leave Management System (ELMS) is a web-based application that helps employees apply for leave and allows administrators to manage leave requests efficiently. This manual provides step-by-step guidance on how to use the system.

How to Access the Web Application

- **Supported Devices:** The system is accessible on desktops, laptops, tablets, and smartphones with an internet browser.
- **Supported Browsers:** Google Chrome, Mozilla Firefox, Microsoft Edge, Safari.
- **Accessing the Application:**
 - Open a web browser and enter the ELMS website URL (provided by the admin).
 - Log in with your credentials to start using the system.

2. Getting Started

- **Launching the Application:** Open the web browser and visit the ELMS website.
- **Home Screen:** The system loads the homepage, where users can log in.

3. Core Features

➤ Apply for Leave

Employees can apply for leave by selecting:

- Leave Type (Casual, Sick, Earned, Restricted, etc.).
- Start and End Date of the leave.
- Reason for Leave (Optional).
- Click Submit to send the request.

➤ View Leave Balance

- The system displays the remaining leave balance for each leave type.

➤ Check Leave Status

- Employees can track the status of applied leave (Pending, Approved, or Rejected).

➤ Admin Features

- **Manage Employees:** Add, update, or remove employee records.
- **Approve/Reject Leave Requests:** Review leave applications and approve or reject them based on availability.

5. Additional Features

- **Email Notifications:** Employees receive real-time email alerts when their leave request is approved or rejected.
- **Role-Based Access:** Employees and admins have different permissions for system access.

4.2 TESTING

Testing is the process of evaluating and verifying that a software application works correctly and meets the required standards. It helps identify bugs, errors, and performance issues before the software is deployed. The main goal of testing is to ensure the software is reliable, secure, and functions as expected. Testing is classified into different types based on the purpose and phase of testing.

4.2.1 UNIT TESTING

Unit testing is a type of software testing where individual components or functions of a software application are tested separately to ensure they work correctly. It focuses on small, isolated parts of the application, such as a function, method, or module, rather than the entire system.

Unit Testing Cases

Module Name	Test Case	Expected Output	Actual Output
Authentication Module	Verify user login with correct credentials	User successfully logs in	Success
Leave Request Module	Apply for leave with valid details	Leave request stored successfully	Request Saved Successfully
	Apply for leave exceeding balance	Error message displayed	Error Displayed
Leave Approval Module	Approve leave request	Leave status updated to 'Approved'	Status Updated
	Reject leave request	Leave status updated to 'Rejected'	Status Updated
Leave Balance Module	Check leave balance after applying leave	Balance updates correctly	Balance Updated
Reports Module	Generate leave history report	Report generated successfully	Report Displayed

5. CONCLUSION AND FUTURE ENHANCEMENT

5.1 CONCLUSION

The Employee Leave Management System (ELMS) is a comprehensive solution designed to streamline leave management processes within an organization. By automating leave applications, approvals, and tracking, the system enhances efficiency, reduces paperwork, and minimizes errors. With features such as real-time leave balance updates, role-based access control, and secure data management, ELMS ensures transparency and fairness in leave distribution. Additionally, the implementation of validation rules prevents misuse and enforces company policies effectively.

Overall, this project improves administrative workflow, enhances employee satisfaction, and provides a scalable and user-friendly platform for managing leaves efficiently. Furthermore, the system's ability to generate detailed reports and analytics helps organizations make data-driven decisions, optimize workforce management, and improve overall productivity. The seamless integration with payroll and HR systems ensures that leave deductions and salary adjustments are automatically managed, reducing manual workload for HR personnel.

ELMS allows organizations to define and modify leave policies based on their specific requirements, making it highly customizable. The system supports multi-user access with different roles, enabling employees, managers, and HR administrators to interact with the platform based on their access levels. To enhance accessibility, the system can be extended to mobile platforms, allowing employees to apply for leave from anywhere.

Automated notifications keep employees informed about leave approvals, rejections, and balance updates, reducing the need for manual follow-ups. The system is designed to be scalable and flexible, making it suitable for small, medium, and large enterprises, with the ability to expand as the organization grows. With a user-friendly interface, ELMS provides an intuitive experience for both employees and administrators, reducing the learning curve and improve operational efficiency, reduce administrative burdens, and create a structured leave management process that promotes transparency and fairness across all levels.

Employee leave management system for managing staff leave in higher institution has successfully been developed. The system was developed in line with three tier architecture

software model and implemented using web based technologies which include CSS, JS, HTML, MySQL. The system serves as improvement in staff management, maintain accuracy, transparency and highlight the need to integrate advance technology in employee record and welfare management in higher institution. The developed system enables the employees in academic institutions to request and track their leave at their own convenient time in timely manner. Superior officers of the institution and administrative department can create leave policies, check transparency and plan activities ahead of time. Further researchers can integrate Unstructured Supplementary Service Data (USSD) into the System to manage employees leave and emergency welfare

5.2 FUTURE ENHANCEMENT

Mobile Application Integration

Developing a mobile-friendly version or dedicated app to allow employees to apply for and track leave on the go.

2. AI-Based Leave Prediction

Implementing AI algorithms to analyze leave patterns and suggest optimal leave planning for employees.

3. Integration with Payroll System

Linking leave records with payroll processing to automate salary deductions for unpaid leaves.

4. Automated Notifications & Reminders

Enhancing the system with automated email/SMS reminders for pending approvals and upcoming leaves.

5. Multi-Level Approval Workflow

Introducing customizable multi-level leave approval workflows based on company hierarchy.

6. Customizable Leave Policies

Allowing HR to configure different leave policies based on employee roles, experience, and departments.

7. Analytics & Reporting

Implementing advanced reporting and analytics to track leave trends, absenteeism, and workforce productivity.

8. Chatbot Assistance

Integrating a chatbot for quick leave queries, balance checks, and application assistance.

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3. Gupta, A. (2017). *Database Management Systems*. McGraw-Hill Education.
4. Mathis, R. L., & Jackson, J. H. (2019). *Human Resource Management*. Cengage Learning.

Websites

- www.oracle.com
- www.ibm.com
- www.labour.gov.in

APPENDIX

(a) TABLE STRUCTURE

DATABASE NAME: elmsdb

TABLE NAME: admin

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	id 	int(11)			No	None		AUTO_INCREMENT	 Change  Drop  More
☐ 2	username	varchar(30)	utf8mb4_general_ci		Yes	NULL			 Change  Drop  More
☐ 3	password	varchar(8)	utf8mb4_general_ci		Yes	NULL			 Change  Drop  More
☐ 4	updatetime	date			No	None			 Change  Drop  More

Table a.1

TABLE NAME: leavecount


Table structure


Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	leavecount_id 	int(11)			No	None		AUTO_INCREMENT	 Change  Drop  More
☐ 2	Emp_id	varchar(20)	utf8mb4_general_ci		Yes	NULL			 Change  Drop  More
☐ 3	leave_count	int(11)			No	None			 Change  Drop  More
☐ 4	create_at	timestamp			No	current_timestamp()		ON UPDATE CURRENT_TIMESTAMP()	 Change  Drop  More

Table a.2

TABLE NAME: employee

Table structure									
Relation view									
#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	emp_id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐ 2	first_name	varchar(30)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 3	last_name	varchar(30)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 4	emailid	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 5	password	varchar(8)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 6	gender	varchar(20)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 7	dob	date			No	None			Change Drop More
☐ 8	department	varchar(30)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 9	address	varchar(200)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 10	city	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 11	country	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 12	phonenumber	int(11)			No	None			Change Drop More
☐ 13	status	varchar(30)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 14	Regdate	date			No	None			Change Drop More

Table a.3

TABLE NAME: leavetype

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	leavetype_id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐ 2	leavetype	varchar(50)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 3	description	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More

Table a.4

TABLE NAME: leaverequest

Table structure		Relation view								
#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action	
☐	1 leaverequest_id	int(11)			No	None		AUTO_INCREMENT	Change	Drop More
☐	2 emp_id	varchar(30)	utf8mb4_general_ci		Yes	NULL			Change	Drop More
☐	3 name	varchar(30)	utf8mb4_general_ci		Yes	NULL			Change	Drop More
☐	4 leavetype	varchar(30)	utf8mb4_general_ci		Yes	NULL			Change	Drop More
☐	5 from_date	date			No	None			Change	Drop More
☐	6 to_date	date			No	None			Change	Drop More
☐	7 description	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change	Drop More
☐	8 postingdate	date			No	None			Change	Drop More
☐	9 adminremark	varchar(50)	utf8mb4_general_ci		Yes	NULL			Change	Drop More
☐	10 adminremarkdate	date			No	None			Change	Drop More
☐	11 leavecount	int(11)			No	None			Change	Drop More
☐	12 status	varchar(50)	utf8mb4_general_ci		No	None			Change	Drop More

Table a.5

(b) SAMPLE CODE

```
<?php

    session_start();

    error_reporting(0);

    include('includes/config.php');

    if (strlen($_SESSION['alogin']) == 0) {

        header('location:index.php');

    } else {

?>


<!DOCTYPE html>

<html lang="en">


<head>


    <!-- Title -->

    <title>Admin | Dashboard</title>


    <meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-
scale=1.0, user-scalable=no" />

    <meta charset="UTF-8">

    <meta name="description" content="Responsive Admin Dashboard Template" />

    <meta name="keywords" content="admin,dashboard" />

    <meta name="author" content="

" />
```



```

<!-- Styles -->

<link type="text/css" rel="stylesheet"
href="../assets/plugins/materialize/css/materialize.min.css" />

<link href="http://fonts.googleapis.com/icon?family=Material+Icons"
rel="stylesheet">

<link href="../assets/plugins/metrojs/MetroJs.min.css" rel="stylesheet">

<link href="../assets/plugins/weather-icons-master/css/weather-icons.min.css"
rel="stylesheet">

<!-- Theme Styles -->

<link href="../assets/css/alpha.min.css" rel="stylesheet" type="text/css" />

<link href="../assets/css/custom.css" rel="stylesheet" type="text/css" />

</head>

<body>

<?php include('includes/header.php'); ?>

<?php include('includes/sidebar.php'); ?>

<main class="mn-inner">

    <div class="">

        <div class="row no-m-t no-m-b">

            <a href="manageemployee.php" target="blank">

                <div class="col s12 m12 l4">

                    <div class="card stats-card">

                        <div class="card-content">

                            <span class="card-title">Totle Regd Employee</span>

                            <span class="stats-counter">

```

```

        <?php
        $sql = "SELECT id from tblemployees";

        $query = $dbh->prepare($sql);

        $query->execute();

        $results = $query->fetchAll(PDO::FETCH_OBJ);

        $sempcount = $query->rowCount();

        ?>

        <span class="counter"><?php echo htmlentities($sempcount);
?></span></span>

```

```

    </div>

```

```

<div class="progress stats-card-progress">

```

```

    <div class="determinate" style="width: 70%"></div>

```

```

</div>

```

```

</div>

```

```

</div>

```

```

</a>

```

```

<a href="manageddepartments.php" target="blank">

```

```

    <div class="col s12 m12 l4">

```

```

        <div class="card stats-card">

```

```

            <div class="card-content">

```

```

                <span class="card-title">Listed Departments </span>

```

```

                <?php

```

```

                $sql = "SELECT id from tbldepartments";

```

```

                $query = $dbh->prepare($sql);

```

```

$query->execute();

$results = $query->fetchAll(PDO::FETCH_OBJ);

$dptcount = $query->rowCount();

?>

<span class="stats-counter"><span class="counter"><?php echo
htmlentities($dptcount); ?></span></span>

</div>

<div class="progress stats-card-progress">

    <div class="determinate" style="width: 70%"></div>

</div>

</div>

</div>

</div>

</a>

<a href="manageleavetype.php" target="blank">

    <div class="col s12 m12 l4">

        <div class="card stats-card">

            <div class="card-content">

                <span class="card-title">Listed leave Type</span>

                <?php

                    $sql = "SELECT id from tblleavetype";

                    $query = $dbh->prepare($sql);

                    $query->execute();

                    $results = $query->fetchAll(PDO::FETCH_OBJ);

                    $leavtypcount = $query->rowCount();

```

```
        ?>

        <span class="stats-counter"><span class="counter"><?php echo
htmlentities($leavtypcount); ?></span></span>
```

```
    </div>

    <div class="progress stats-card-progress">

        <div class="determinate" style="width: 70%"></div>

    </div>

</div>

</div>

</a>
```

```
<a href="leaves.php" target="blank">

    <div class="col s12 m12 l4">

        <div class="card stats-card">

            <div class="card-content">

                <span class="card-title">Total Leaves</span>

                <?php

                $sql = "SELECT id from tblleaves";

                $query = $dbh->prepare($sql);

                $query->execute();

                $results = $query->fetchAll(PDO::FETCH_OBJ);

                $totalleaves = $query->rowCount();

                ?>
```

```
        <span class="stats-counter"><span class="counter"><?php echo  
htmlentities($totalleaves); ?></span></span>
```

```
    </div>
```

```
    <div class="progress stats-card-progress">
```

```
        <div class="success" style="width: 70%"></div>
```

```
    </div>
```

```
</div>
```

```
</div>
```

```
</a>
```

```
<a href="approvedleave-history.php" target="blank">
```

```
    <div class="col s12 m12 l4">
```

```
        <div class="card stats-card">
```

```
            <div class="card-content">
```

```
                <span class="card-title">Approved Leaves</span>
```

```
                <?php
```

```
                    $sql = "SELECT id from tblleaves where Status=1";
```

```
                    $query = $dbh->prepare($sql);
```

```
                    $query->execute();
```

```
                    $results = $query->fetchAll(PDO::FETCH_OBJ);
```

```
                    $approvedleaves = $query->rowCount();
```

```
                ?>
```

```
                <span class="stats-counter"><span class="counter"><?php echo  
htmlentities($approvedleaves) ?? " "; ?></span></span>
```

```

</div>

<div class="progress stats-card-progress">

    <div class="success" style="width: 70%"></div>

</div>

</div>

</div>

</a>

```

```

<a href="pending-leavehistory.php" target="blank">

<div class="col s12 m12 l4">

<div class="card stats-card">

<div class="card-content">

    <span class="card-title">New Leave Applications</span>

    <?php

    $sql = "SELECT id from tblleaves where Status=0";

    $query = $dbh->prepare($sql);

    $query->execute();

    $results = $query->fetchAll(PDO::FETCH_OBJ);

    $newleaveapplications = $query->rowCount();

    ?>

    <span class="stats-counter"><span class="counter"><?php echo
htmlentities($newleaveapplications); ?></span></span>

```

```

        </div>

        <div class="progress stats-card-progress">

            <div class="success" style="width: 70%"></div>

        </div>

    </div>

</div>

</a>

</div>

<div class="row no-m-t no-m-b">

    <div class="col s15 m12 l12">

        <div class="card invoices-card">

            <div class="card-content">

                <span class="card-title">Latest Leave Applications</span>

                <table id="example" class="display responsive-table ">

                    <thead>

                        <tr>

                            <th>#</th>

                            <th width="200">Employee Name</th>

                            <th width="120">Leave Type</th>

                            <th width="180">Posting Date</th>

                            <th>Status</th>

```

```

        <th align="center">Action</th>

    </tr>

</thead>

<tbody>

<?php

    $sql = "SELECT tblleaves.id as lid, tblemployees.FirstName, tblemployees.LastName,
tblemployees.EmpId,

        tblemployees.id, tblleaves.LeaveType, tblleaves.PostingDate, tblleaves.Status

    FROM tblleaves

    JOIN tblemployees ON tblleaves.empid = tblemployees.id

    ORDER BY lid DESC LIMIT 6";

    $query = $dbh->prepare($sql);

    $query->execute();

    $results = $query->fetchAll(PDO::FETCH_OBJ);

    echo $result;

    $cnt = 1;

    if ($query->rowCount() > 0) {

        foreach ($results as $result

            ?>

            <tr>

                <td><b><?php echo htmlentities($cnt); ?></b></td>

                <td>

                    <a href="editemployee.php?empid=<?php echo htmlentities($result->id); ?>"
target="_blank">

```



```

        <?php echo htmlentities($result->FirstName . " " . $result->LastName); ?>

        (<?php echo htmlentities($result->EmpId); ?>)

    </a>

</td>

<td><?php echo htmlentities($result->LeaveType); ?></td>

<td><?php echo htmlentities($result->PostingDate); ?></td>

<td>

    <?php

    $stats = $result->Status;

    if ($stats == 1) {

        echo '<span style="color: green">Approved</span>';

    } elseif ($stats == 2) {

        echo '<span style="color: red">Not Approved</span>';

    } else {

        echo '<span style="color: blue">Waiting for approval</span>';

    }

    ?>

</td>

<td>

    <a href="leave-details.php?leaveid=<?php echo htmlentities($result->lid); ?>"

        class="waves-effect waves-light btn blue m-b-xs">View Details</a>

</td>

</tr>

<?php

```

```

        $cnt++;
    }
}
?>
</tbody>

        </table>

    </div>

</div>

</div>

</div>

</div>

</main>

</div>

<!-- Javascripts -->

<script src="../../assets/plugins/jquery/jquery-2.2.0.min.js"></script>

<script src="../../assets/plugins/materialize/js/materialize.min.js"></script>

<script src="../../assets/plugins/material-
preloader/js/materialPreloader.min.js"></script>

<script src="../../assets/plugins/jquery-blockui/jquery.blockui.js"></script>

<script src="../../assets/plugins/waypoints/jquery.waypoints.min.js"></script>

<script src="../../assets/plugins/counter-up-master/jquery.counterup.min.js"></script>

<script src="../../assets/plugins/jquery-sparkline/jquery.sparkline.min.js"></script>

<script src="../../assets/plugins/chart.js/chart.min.js"></script>

```

```
<script src="../assets/plugins/flot/jquery.flot.min.js"></script>

<script src="../assets/plugins/flot/jquery.flot.time.min.js"></script>

<script src="../assets/plugins/flot/jquery.flot.symbol.min.js"></script>

<script src="../assets/plugins/flot/jquery.flot.resize.min.js"></script>

<script src="../assets/plugins/flot/jquery.flot.tooltip.min.js"></script>

<script src="../assets/plugins/curvedlines/curvedLines.js"></script>

<script src="../assets/plugins/peity/jquery.peity.min.js"></script>

<script src="../assets/js/alpha.min.js"></script>

<script src="../assets/js/pages/dashboard.js"></script>

</body>

</html>

<?php
```

(c) INPUT FORMS

The screenshot shows the login interface of the Employee Leave Management System (ELMS). The header bar is teal and contains the text "ELMS | EMPLOYEE LEAVE MANAGEMENT SYSTEM". On the left, a sidebar lists "Employee Login" and "Admin Login". The main content area has a heading "WELCOME TO EMPLOYEE LEAVE MANAGEMENT SYSTEM" and a central "EMPLOYEE LOGIN" form. The form includes input fields for "Email Id" and "Password", and a teal "SIGN IN" button. The footer shows "ELMS ©".

Figure c.1

The screenshot shows the employee dashboard of the Employee Leave Management System (ELMS). The header bar is teal and contains the text "ELMS | EMPLOYEE". On the left, a sidebar shows the user profile "Siva L EMP405" and a menu with "Dashboard", "My Profiles", "Change Password", "Leaves", and "Sign Out". The main content area has a heading "WELCOME TO EMPLOYEE LEAVE MANAGEMENT SYSTEM" and a "BALANCE LEAVES THIS YEAR" card displaying the value "24". The footer shows "ELMS ©".

Figure c.2

ELMS | EMPLOYEE

Siva L

EMP405

Dashboard

My Profiles

Change Password

Leaves >

Sign Out

APPLY FOR LEAVE

Apply for Leave

Employee ID : EMP405

Name

Select leave type...

From Date
dd-mm-yyyy

To Date
dd-mm-yyyy

Description

APPLY

ELMS ©

Figure c.3

ELMS | ADMIN

Admin

Dashboard

Department >

Leave Type >

Employees >

Leave Management >

Change Password

Sign Out

ADD LEAVE TYPE

Leave Type

Description

ADD

ELMS ©

Figure c.4

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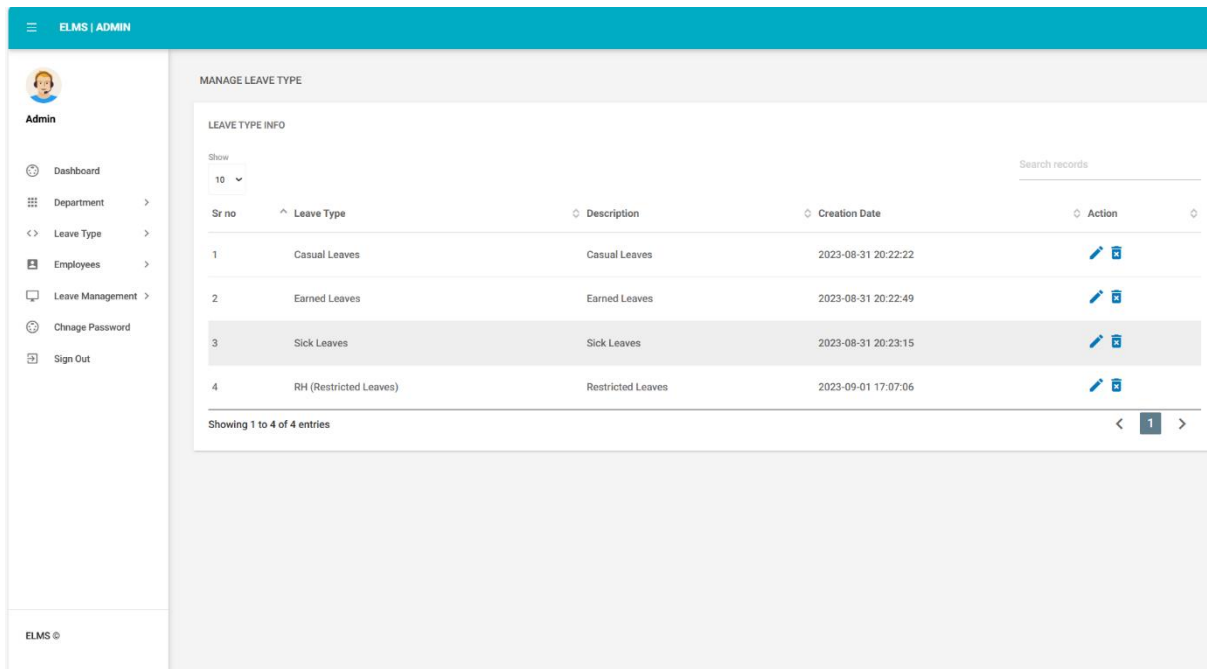


Figure c.5

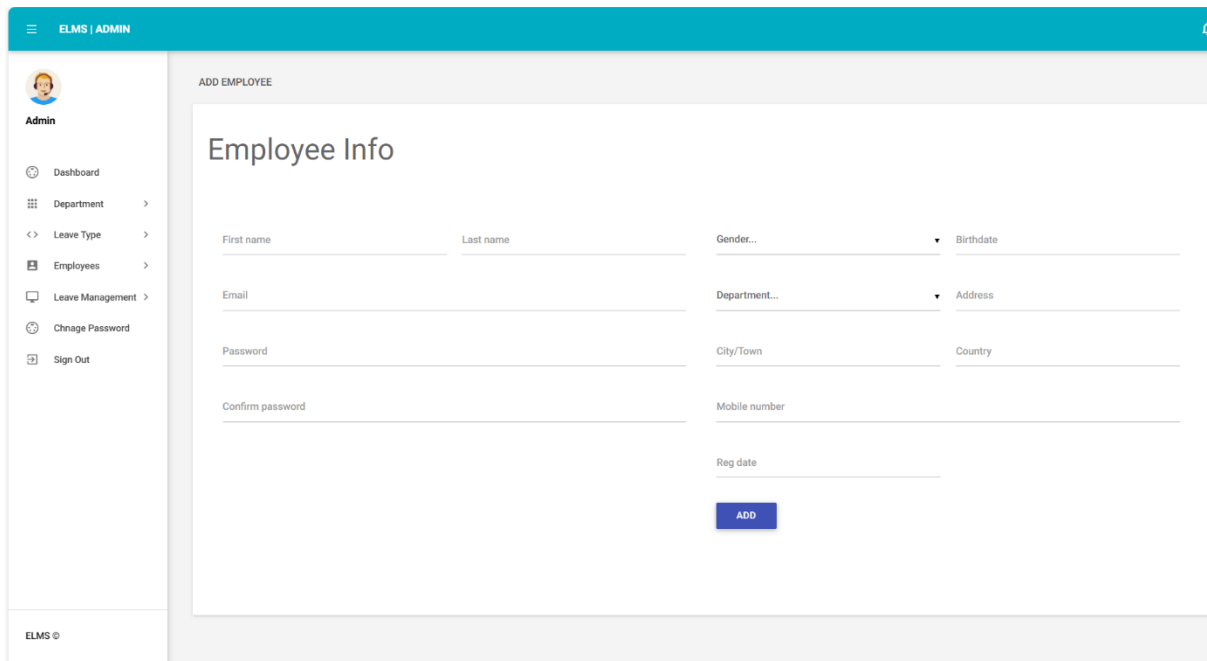


Figure c.6

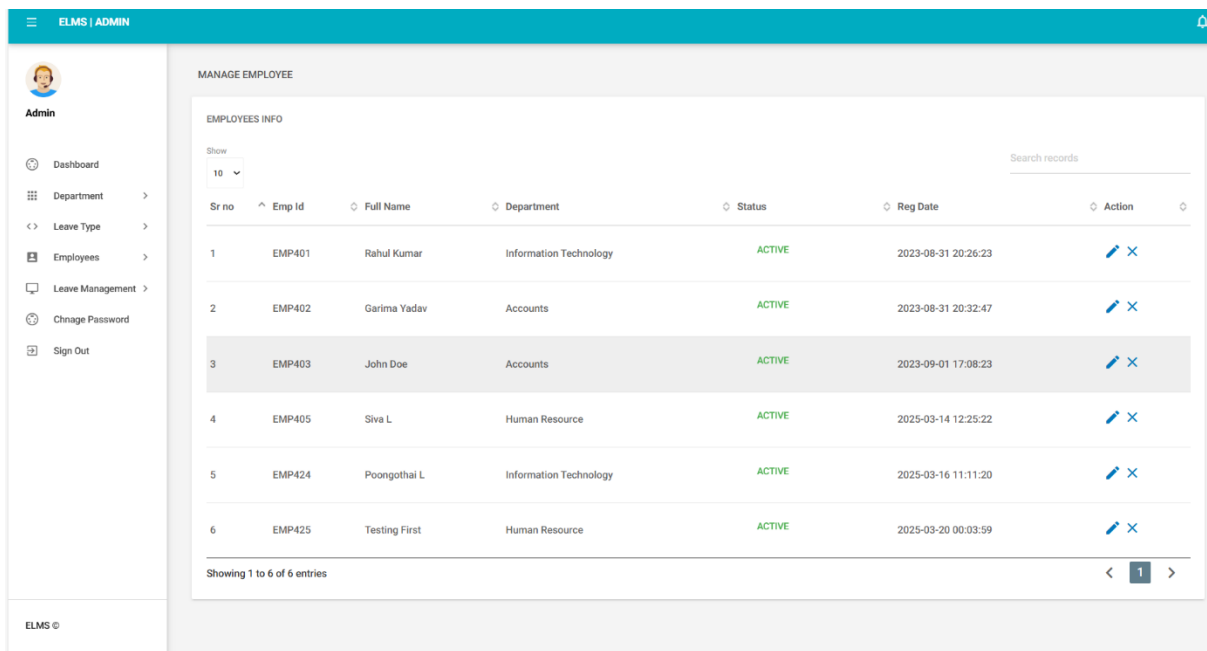


Figure c.7

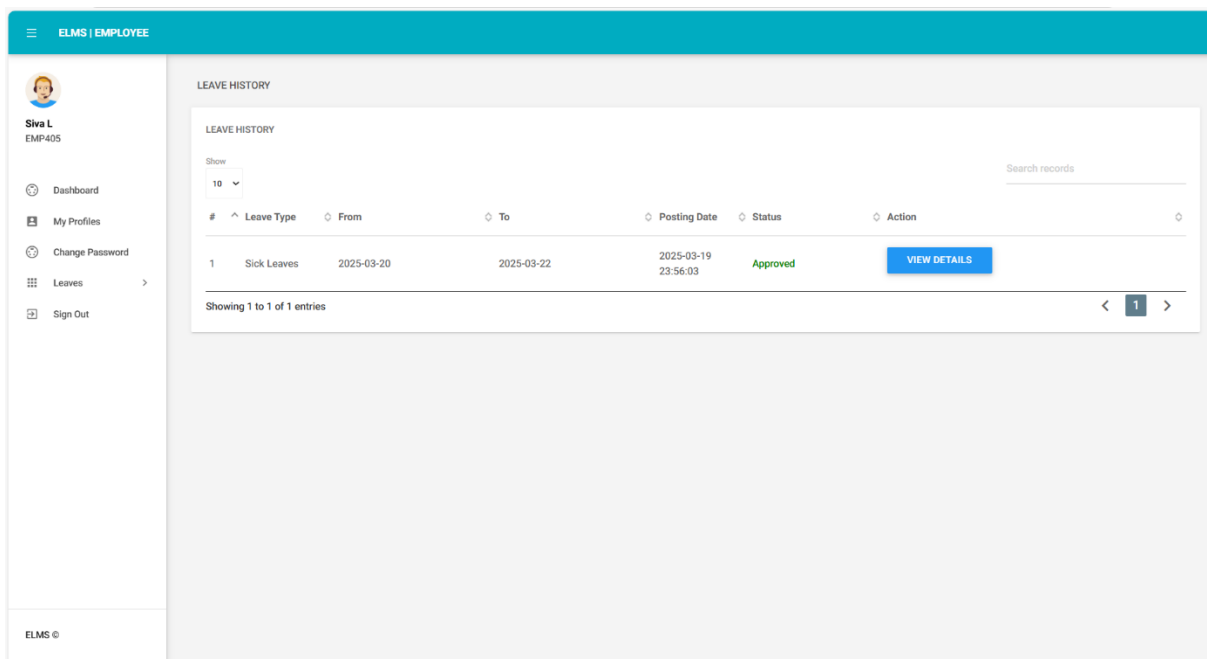


Figure c.8

(d) OUTPUT FORMS

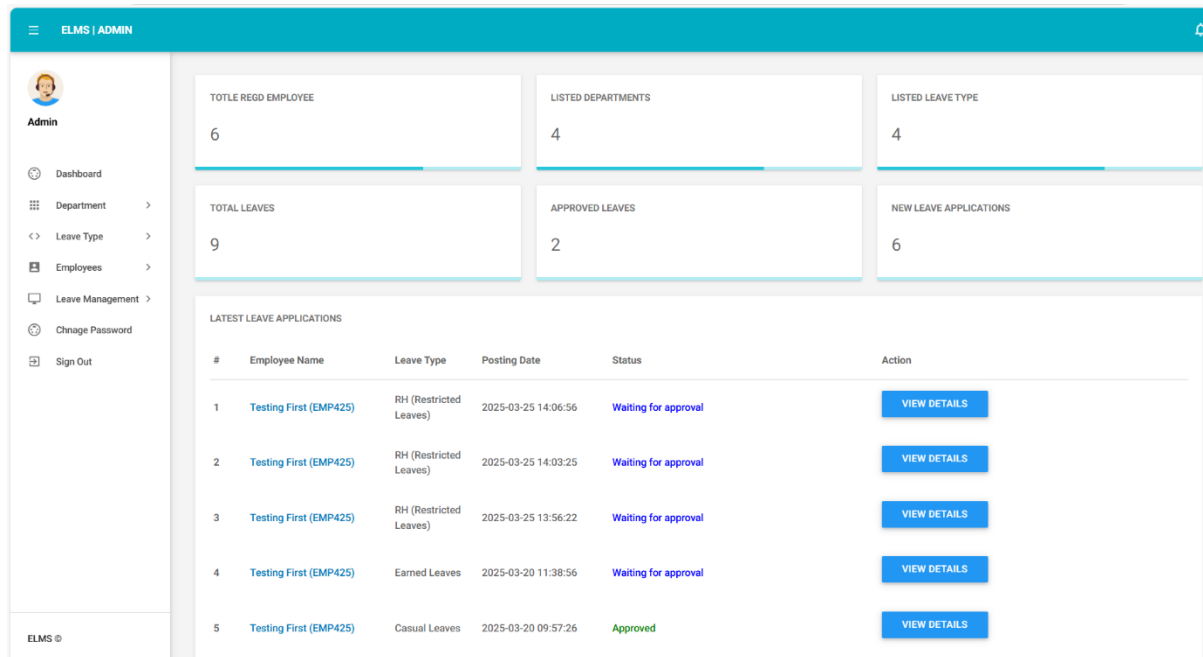


Figure d.1

ELMS ADMIN						
Admin Dashboard Department > <> Leave Type > Employees > Leave Management > All Leaves Pending Leaves Approved Leaves Not Approved Leaves Change Password Sign Out ELMS ©	PENDING LEAVE HISTORY					
	LEAVE HISTORY					
	#	Employee Name	Leave Type	Posting Date	Status	Action
	1	Testing First(EMP425)	RH (Restricted Leaves)	2025-03-25 14:06:56	waiting for approval	<button>VIEW DETAILS</button>
	2	Testing First(EMP425)	RH (Restricted Leaves)	2025-03-25 14:03:25	waiting for approval	<button>VIEW DETAILS</button>
	3	Testing First(EMP425)	RH (Restricted Leaves)	2025-03-25 13:56:22	waiting for approval	<button>VIEW DETAILS</button>
	4	Testing First(EMP425)	Earned Leaves	2025-03-20 11:38:56	waiting for approval	<button>VIEW DETAILS</button>
	5	John Doe(EMP403)	Sick Leaves	2025-03-16 10:54:51	waiting for approval	<button>VIEW DETAILS</button>
	6	Rahul Kumar(EMP401)	Casual Leaves	2023-08-31 20:36:21	waiting for approval	<button>VIEW DETAILS</button>

Figure d.2

ELMS ADMIN						
Admin Dashboard Department > <> Leave Type > Employees > Leave Management > All Leaves Pending Leaves Approved Leaves Not Approved Leaves Change Password Sign Out ELMS ©	LEAVE DETAILS					
	LEAVE DETAILS					
	SUCCESS : Leave updated Successfully					
	Employee Name :	Testing First	Emp Id :	EMP425	Gender :	
	Emp Email Id :	gokulbtech07@gmail.com	Emp Contact No. :	7877677777		
	Leave Type :	RH (Restricted Leaves)	Leave Date . :	From 2025-03-25 to 2025-03-28	Posting Date	2025-03-25 14:06:56
	Employee Leave Description :	Testing				
	leave Status :	Approved				
	Admin Remark:	your leave has been approved				
	Admin Action taken date :	2025-03-28 12:59:03				
	Balance Leave Count:	12				

Figure d.3